

2005

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 15, 17, and 21

Total marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–36

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

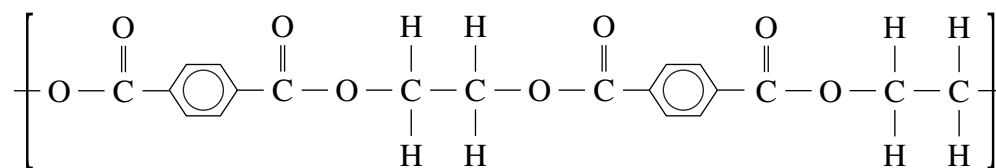
A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1** Which of the following conditions would produce a radioactive isotope?
- (A) Too many atoms in the sample provided
 - (B) Too many protons and neutrons in the atom
 - (C) Too many electrons in the outer shell of the atom
 - (D) Too many electrons for the number of neutrons in the atom
- 2** Which of the following makes cellulose a potential raw material for the petrochemical industry?
- (A) It is available in many parts of the world.
 - (B) It is light in weight and easy to transport.
 - (C) It is an organic polymer based on carbon.
 - (D) It is a polymer that undergoes combustion.
- 3** The heat of combustion of butan-1-ol is 2676 kJ mol^{-1} .
What is the value of the heat of combustion in kJ g^{-1} ?
- (A) 30.41
 - (B) 36.10
 - (C) 44.60
 - (D) 47.79

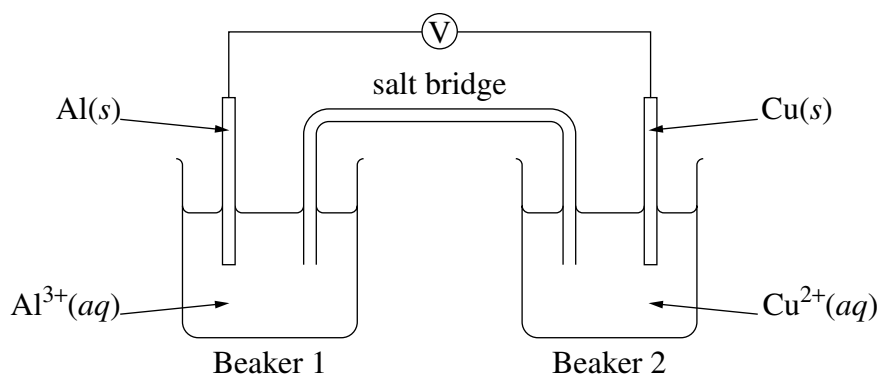
- 4 Terylene (polyester) is a condensation polymer. Part of the structure of the polymer is shown.



What are the two monomers that form this polymer?

	<i>Monomer 1</i>	<i>Monomer 2</i>
(A)	$\text{H}-\text{C}_6\text{H}_4-\text{H}$	$\text{HO}-\underset{\text{H}}{\underset{\text{H}}{\text{C}}}-\underset{\text{H}}{\underset{\text{H}}{\text{C}}}-\text{OH}$
(B)	$\text{HO}-\underset{\text{H}}{\text{C}}=\underset{\text{H}}{\text{C}}-\text{OH}$	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_4-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
(C)	$\text{H}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	$\text{H}-\text{C}_6\text{H}_4-\text{H}$
(D)	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_4-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$	$\text{HO}-\underset{\text{H}}{\underset{\text{H}}{\text{C}}}-\underset{\text{H}}{\underset{\text{H}}{\text{C}}}-\text{OH}$

- 5 An electrochemical cell is set up as shown in the diagram.



What are two observations for this electrochemical cell?

	<i>Observation 1</i>	<i>Observation 2</i>
(A)	A reading was shown on the voltmeter	In Beaker 2 the solution became a darker blue
(B)	In Beaker 2 the blue solution faded	A reddish precipitate formed on the copper electrode
(C)	A grey precipitate formed on the aluminium electrode	In Beaker 2 the solution became a darker blue
(D)	Electrons moved through the voltmeter	A reddish precipitate formed on the copper electrode

- 6 Which of the following is a common use for ethyl pentanoate?

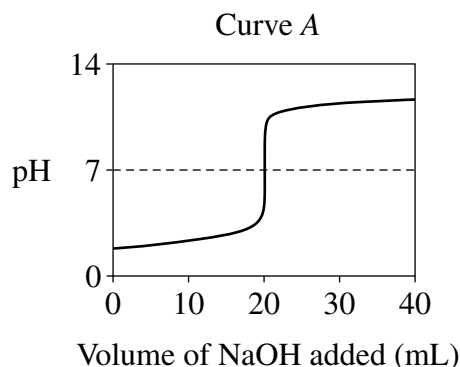
- (A) Flavouring
- (B) Fuel
- (C) Indicator
- (D) Solvent

- 7 A series of observations was made about a first-hand investigation.
- A cup of carbonated soft drink was poured from a newly-opened, chilled bottle.
 - A lot of bubbles were observed in the cup.
 - The cup was then left on the kitchen bench for several hours.
 - After this time, it was observed that there were no bubbles visible in the cup.

Why were there no bubbles visible in the cup after several hours?

- (A) The bubbles of carbon dioxide had dissolved into the soft drink.
 - (B) The carbon dioxide had been neutralised by the water in the soft drink.
 - (C) Carbon dioxide is less soluble at higher temperatures and lower pressures.
 - (D) Carbon dioxide is more soluble at higher temperatures and lower pressures.
- 8 What would be the pH of a 0.1 mol L^{-1} solution of sulfuric acid?
- (A) Less than 1.0
 - (B) Exactly 1.0
 - (C) Between 1.0 and 7.0
 - (D) Greater than 7.0
- 9 Which of the following pairs would form a buffer solution?
- (A) $\text{HCl}(aq) / \text{Cl}^-(aq)$
 - (B) $\text{H}_2\text{PO}_4^-(aq) / \text{PO}_4^{3-}(aq)$
 - (C) $\text{H}_2\text{SO}_4(aq) / \text{HSO}_4^-(aq)$
 - (D) $\text{CH}_3\text{COOH}(aq) / \text{CH}_3\text{COO}^-(aq)$

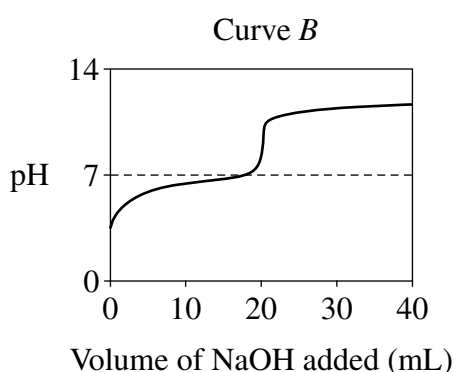
- 10 A titration was conducted by adding NaOH from a teflon-coated burette to HCl in a conical flask. The pH in the flask was recorded during the titration and Curve A was produced.



The table shows appropriate indicators used to identify the equivalence point in titrations. For NaOH and HCl the appropriate indicator is bromothymol blue.

<i>Indicator</i>	<i>Acidic colour</i>	<i>Range of colour change</i>	<i>Basic colour</i>
Methyl orange	red	3.1 – 4.4	yellow
Methyl red	red	4.4 – 6.2	yellow
Bromothymol blue	yellow	6.0 – 7.6	blue
Cresolphthalein	colourless	8.1 – 9.7	red
Alizarin yellow	yellow	10.1 – 12.0	red

A second titration was conducted by adding NaOH to a different acid. The pH in the flask was recorded during the titration and Curve B was produced.



What is the appropriate indicator for Curve B using the table?

- (A) Methyl orange
- (B) Methyl red
- (C) Cresolphthalein
- (D) Alizarin yellow

11 Which of the following substances is best analysed by atomic absorption spectroscopy (AAS)?

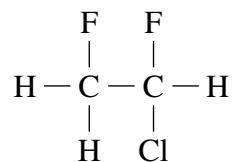
- (A) Calcium
- (B) Iodine
- (C) Nitrogen
- (D) Silicon

12 A student performed a sampling technique as part of a first-hand investigation on water quality and repeated the sampling technique several times.

What aspect of the experiment was improved by repeating the procedure?

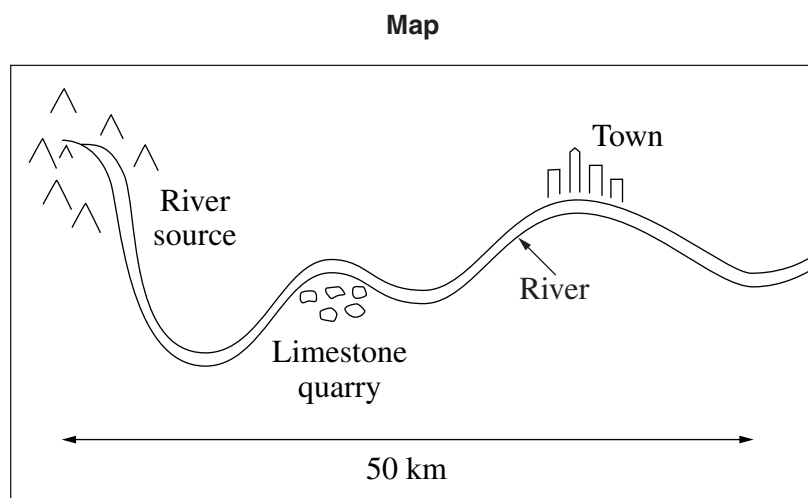
- (A) Accuracy
- (B) Reliability
- (C) Safety
- (D) Validity

13 What is the name of the following compound?



- (A) 1,2-difluoro-1-chloroethane
- (B) 1,2-difluoro-2-chloroethane
- (C) 1-chloro-1,2-difluoroethane
- (D) 2-chloro-1,2-difluoroethane

- 14 Water quality analyses were performed at different sites on a river shown on the map.



The table shows the results of the tests performed at the different sites.

Results of tests performed			
Site	Dissolved oxygen (ppm)	Biochemical oxygen demand (ppm)	pH
1	3	145	5.2
2	10	2	6.9
3	7	12	7.4

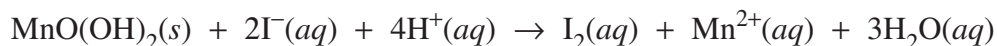
Where was each set of tests performed?

	River source	Limestone quarry	Town
(A)	Site 2	Site 1	Site 3
(B)	Site 1	Site 2	Site 3
(C)	Site 1	Site 3	Site 2
(D)	Site 2	Site 3	Site 1

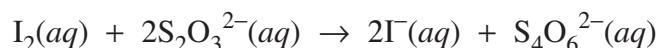
- 15 The Winkler method is used to determine the amount of dissolved oxygen in a sample. In this procedure, oxygen reacts with Mn^{2+} under alkaline conditions to produce a precipitate of $\text{MnO}(\text{OH})_2$.



The precipitate is then dissolved in acid and reacted with iodide, forming iodine and Mn^{2+} .



Finally, the amount of iodine produced is determined by reaction with thiosulfate.



When a sample of water was analysed using the Winkler method, a total of 0.60 mol of thiosulfate was used in the reaction.

How many moles of oxygen were present in the original sample?

- (A) 0.15
- (B) 0.30
- (C) 0.60
- (D) 1.20

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (5 marks)

During your practical work you performed a first-hand investigation to distinguish between an alkene and the corresponding alkane.

- (a) Name the alkene used in your investigation. 1

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- (b) Identify a potential hazard in your investigation, and outline how you addressed this hazard. 2

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- (c) Outline the procedure you used for your first-hand investigation. 2

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Question 17 (3 marks)

The heat of combustion of ethanol is 1367 kJ mol^{-1} . In a first-hand investigation to determine the heat of combustion of ethanol, the experimental value determined differed from the theoretical value.

- (a) Identify a reason for this difference. **1**

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- (b) Calculate the theoretical mass of ethanol required to heat 200 mL of water from 21.0°C to 45.0°C . **2**

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Question 18 (4 marks)

Assess the impact of the recent development of a named biopolymer on society and the environment. **4**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 19 (4 marks)

You have studied one of the cells shown below.



Cell X



Cell Y

Choose ONE of the cells and answer parts (a) and (b).

- (a) State ONE environmental impact associated with the cell you have chosen. 1

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- (b) Describe the chemistry of the cell you have chosen. 3

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 21** (5 marks)

Analyse how knowledge of the composition and properties of acids has led to changes in the definition of acids.

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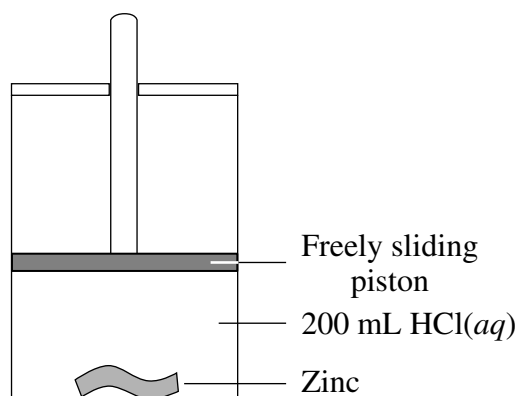
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Question 22 (7 marks)

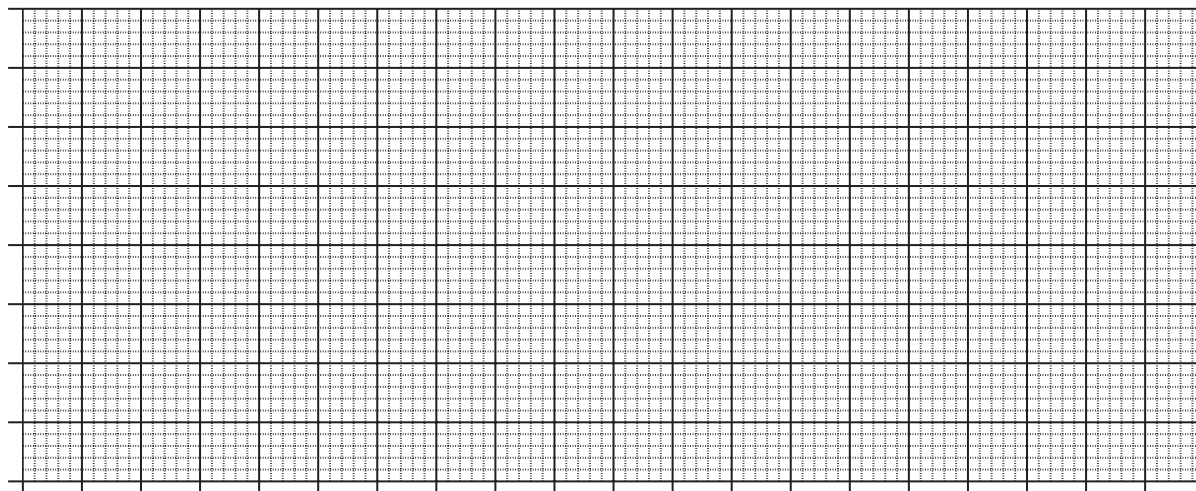
A student carried out a first-hand investigation to identify the relationship between the mass of a metal used in a reaction, and the volume of gas produced. The first-hand investigation was carried out at 25°C, and 100 kPa. In each experiment, 200 mL of hydrochloric acid was added to some zinc, and the volume of gas produced was recorded. The diagram shows the equipment used, and the table contains the student's results.

<i>Mass of zinc (g)</i>	<i>Volume of gas collected at 25°C and 100 kPa (mL)</i>
0.12	45
0.33	125
0.56	115
0.83	315
0.96	365
1.22	380
1.64	380
1.93	380



(a) Graph the results from these experiments.

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Question 22 continues on page 19

Question 22 (continued)

- (b) Predict the volume of gas that would be produced in this experiment if 3.00 g of zinc had been used. Justify your answer. 2

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- (c) Calculate the theoretical volume of gas produced at 25°C and 100 kPa by the reaction of 0.56 g of zinc with the 200 mL of hydrochloric acid. 2

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Question 23 (2 marks)

- (a) What causes incomplete combustion? 1

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- (b) Write a balanced equation for the incomplete combustion of methane. 1

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Question 24 (5 marks)

An antacid tablet is known to contain calcium carbonate (CaCO_3). To determine the mass of calcium carbonate in the tablet, the following procedure was used.

- The tablet was crushed and then placed in a beaker.
- A pipette was used to add 25.0 mL of 0.600 mol L^{-1} hydrochloric acid to the crushed tablet in the beaker.
- Once the reaction between the calcium carbonate and hydrochloric acid had stopped, phenolphthalein indicator was added to the reaction mixture.
- A teflon-coated burette was then used to add 0.100 mol L^{-1} sodium hydroxide to the beaker to neutralise the excess hydrochloric acid.
- The phenolphthalein changed from colourless to pink after 14.2 mL of the sodium hydroxide solution had been added.

- (a) Write a balanced chemical equation for the reaction that occurred between the calcium carbonate in the tablet and the hydrochloric acid. **1**

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- (b) How many moles of hydrochloric acid were added to the tablet? **1**

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- (c) Calculate the mass of calcium carbonate in the original antacid tablet. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 25 (6 marks)

A student collected a 500 mL sample of water from a local creek for analysis. It was filtered and the filtrate evaporated to dryness. The following data were collected.

Mass of filter paper	0.16 g
Mass of filter paper and solid	0.19 g
Mass of evaporating basin	45.33 g
Mass of basin and solid remaining	45.59 g

- (a) Calculate the percentage of total dissolved solids in the creek sample. 2

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- (b) It is suspected that lead has contaminated the creek. Describe a chemical test that could be carried out on the water sample to determine the presence of lead. 2

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- (c) The concentrations of ions in substances used by society need to be monitored. Justify this statement with reference to ONE ion you have studied. 2

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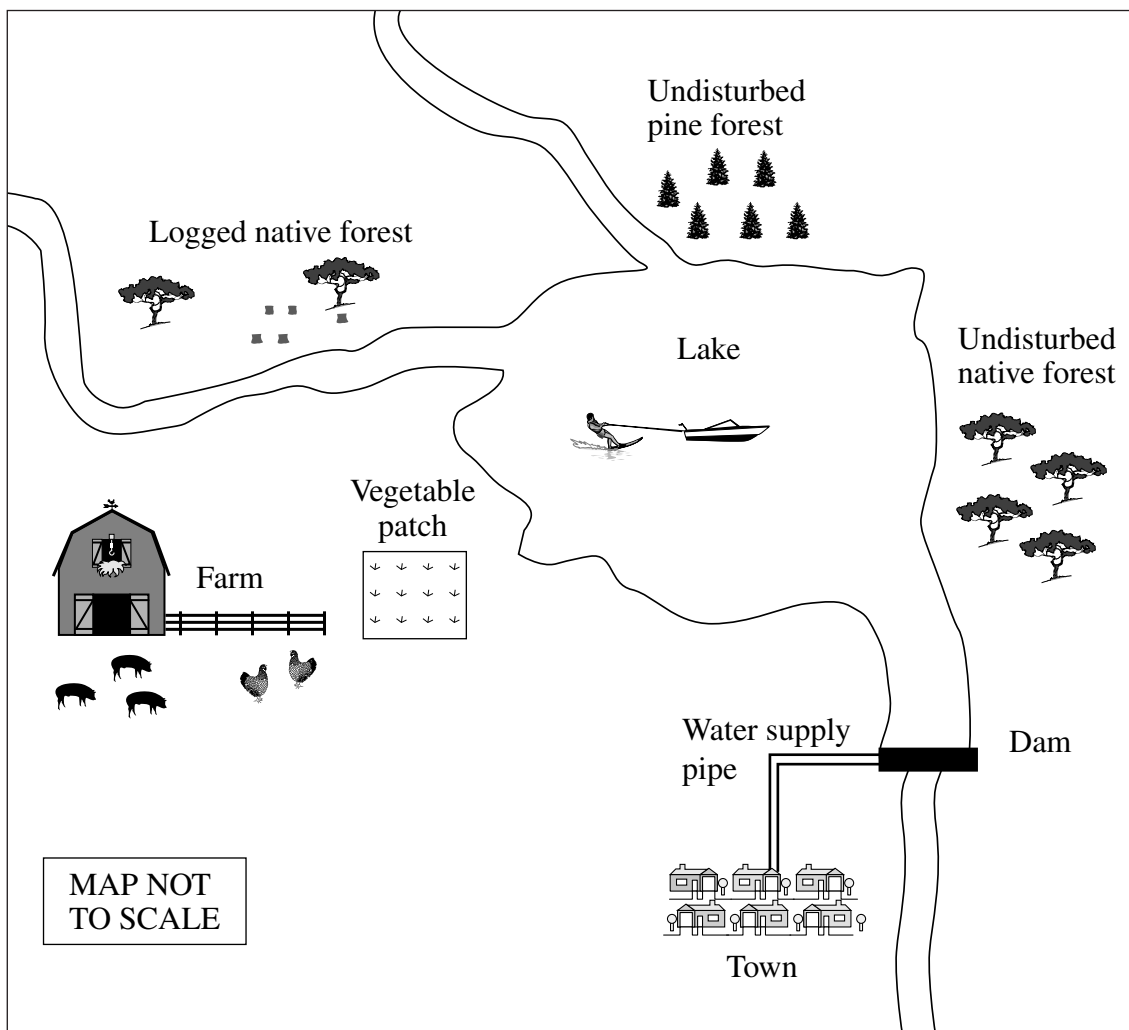
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Question 26 (7 marks)

The map shows the catchment for a town water supply.



Question 26 continues on page 23

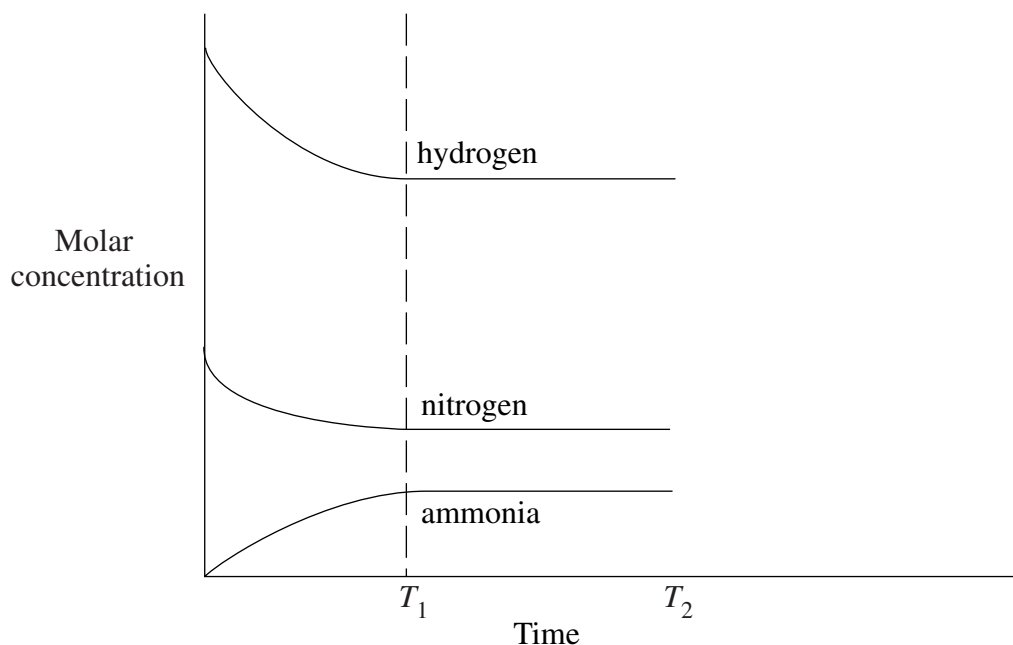
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Question 27 (5 marks)

The graph shows the variation in concentrations of reactants and product with time for the Haber process.



- (a) State why the concentrations of reactants and product do not change between T_1 and T_2 . **1**

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- (b) At time T_2 the volume of the reaction vessel was reduced.

- (i) Sketch on the graph how the concentrations of reactants and product would change after the volume was reduced. **2**

- (ii) Explain the changes shown on your graph. **2**

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Industrial Chemistry	26–27
Question 29 Shipwrecks, Corrosion and Conservation	28–29
Question 30 The Biochemistry of Movement	30–32
Question 31 The Chemistry of Art	33–34
Question 32 Forensic Chemistry	35–36

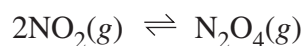
Question 28 — Industrial Chemistry (25 marks)

- (a) Sulfuric acid is an important industrial chemical. One of the intermediate products in the synthesis of sulfuric acid is $\text{H}_2\text{S}_2\text{O}_7$.

(i) What is the name of $\text{H}_2\text{S}_2\text{O}_7$? **1**

(ii) With reference to the properties of sulfuric acid, describe safety precautions necessary for its transport. **3**

- (b) Nitrogen dioxide forms an equilibrium mixture with dinitrogen tetroxide as shown.



At 100°C , K for this reaction is 2.08.

At 25°C , a 1.00 L vessel initially contained 0.132 mol of $\text{NO}_2(\text{g})$. Once equilibrium had been established, there was 0.0400 mol of $\text{N}_2\text{O}_4(\text{g})$ in the vessel.

(i) Explain the effects of the addition of a catalyst and an increase in pressure on the yield of N_2O_4 in this reaction when carried out at 25°C . **2**

(ii) Calculate the equilibrium constant for this reaction at 25°C , and account for any difference from the K value at 100°C . **4**

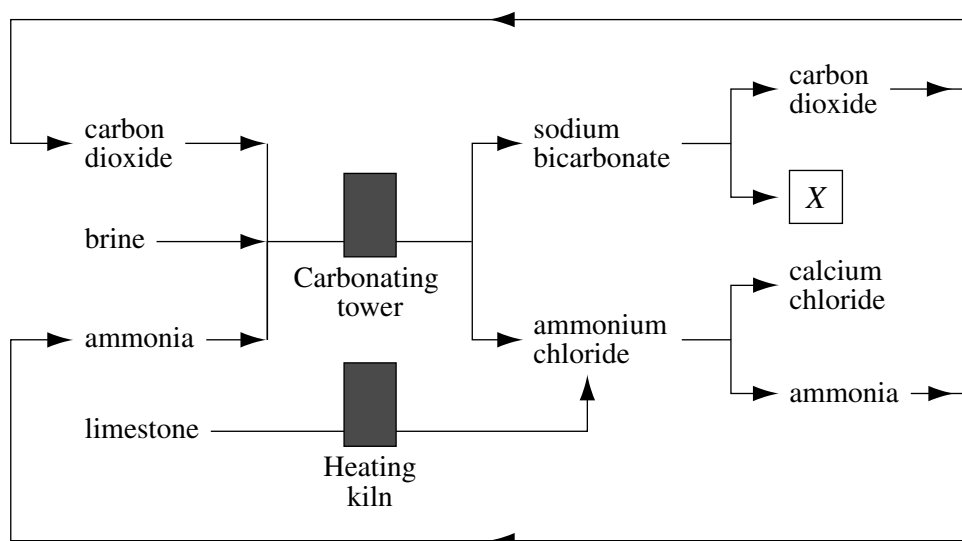
- (c) Advances in chemistry have impacted on the development of technologies. **7**

Discuss this statement by analysing changes that have occurred in industrial methods for the production of sodium hydroxide.

Question 28 continues on page 27

Question 28 (continued)

(d) The diagram summarises the Solvay process.



(i) Identify compound X. 1

(ii) During the Solvay process, ammonia is recovered. 2

Outline the chemistry involved in the recovery of ammonia, and suggest a reason why this recovery is desirable.

(iii) Identify possible environmental issues that may arise from the Solvay process, and evaluate methods of dealing with those environmental issues. 5

End of Question 28

Question 29 — Shipwrecks, Corrosion and Conservation (25 marks)

- (a) (i) Reactive metals such as aluminium and zinc form a strongly adhering oxide coating as a result of the reaction with oxygen. **1**

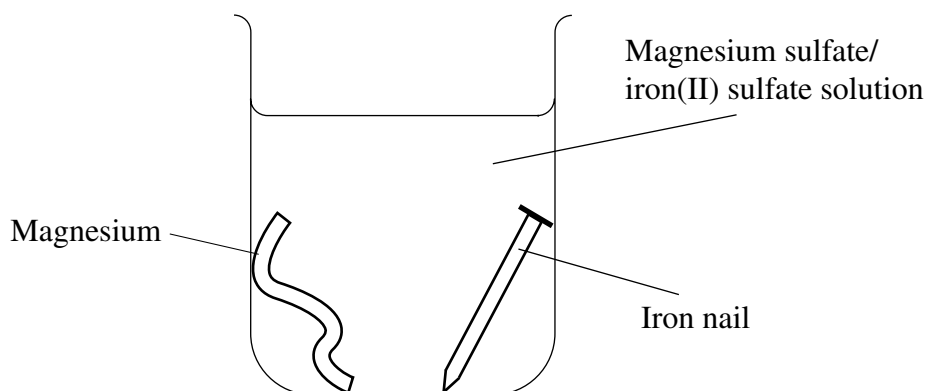
What name is given to these metals?

- (ii) The hulls of modern ships contain steels that have lower levels of carbon and sulfur than was used in the past. **3**

Compare the properties and uses of TWO types of steel that you have studied.

- (b) (i) Outline ONE factor that influences the rate of corrosion of iron in deep-water wrecks. **2**

- (ii) A student set up a laboratory experiment in which a beaker contained a strip of magnesium, an iron nail, and a solution of iron(II) sulfate and magnesium sulfate. This was left for several days. **4**



Describe what would happen in this experiment. Include relevant chemical equations in your answer.

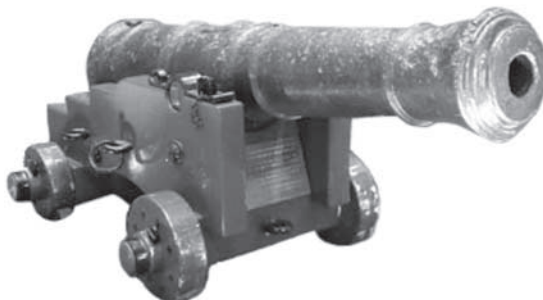
Question 29 continues on page 29

Question 29 (continued)

- (c) Advances in chemistry have impacted on the development of technologies. 7

Discuss this statement by analysing current methods of corrosion protection for ships.

- (d) An iron cannon was recovered from a wreck on the Great Barrier Reef.



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The process used to restore the cannon was as follows:

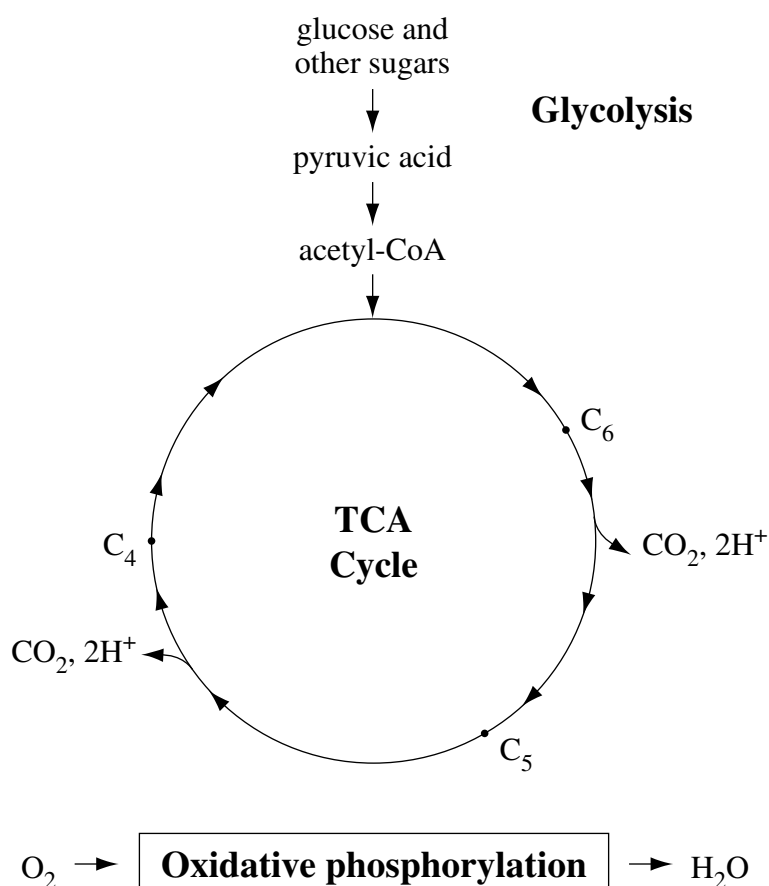
- Step 1: The cannon was placed in dilute sodium hydroxide for several weeks. During this period, the solution was replaced by fresh dilute sodium hydroxide solution at regular intervals.
- Step 2: Coral that had grown on the cannon was carefully chipped away.
- Step 3: Electrolysis was performed on the cannon.
- Step 4: The restored cannon was then coated in wax.

- (i) Why was the restored cannon coated in wax? 1
- (ii) Explain the purpose of Step 1 in the restoration of the iron cannon. 2
- (iii) Describe and justify the procedure for Step 3. Include relevant chemical equations in your answer. 5

End of Question 29

Question 30 — The Biochemistry of Movement (25 marks)

- (a) (i) Draw the structure of glycerol. 1
- (ii) Explain the viscosity of glycerol and its solubility in water. 3
- (b) The diagram below summarises some of the steps involved in respiration.

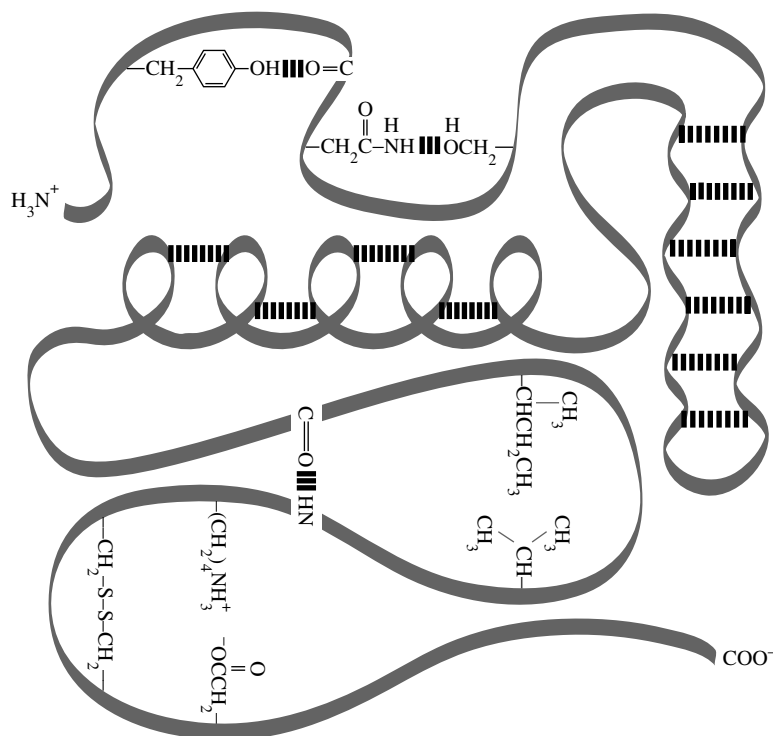


- (i) Compare the energy output from glycolysis with the total energy output from respiration. 2
- (ii) Explain the role of oxidation and reduction in respiration. 4

Question 30 continues on page 31

Question 30 (continued)

- (c) A diagram illustrating the structure of a protein is shown.



Describe how chemical features affect the structure and shape of proteins, and assess the importance of shape in the current model of enzyme function.

7

Question 30 continues on page 32

Question 30 (continued)

- (d) An athletics coach persuaded his runners to drink one of two sports drinks immediately before a race to boost performance. The composition of the two drinks is shown in the table. The coach believes that the sports drinks will give the runners extra energy during both a sprint race and a marathon.

<i>Name of sports drink</i>	<i>Carbohydrate content (%)</i>	<i>Carbohydrate composition</i>	<i>Sodium (mg per 500 mL)</i>
Athleaid	6	sucrose glucose fructose	225
Energade	8	fructose maltodextrin*	130

*Maltodextrin is a polymer composed of many glucose molecules bonded together.

- (i) Define the term *carbohydrate*. 1
- (ii) Describe the chemical process involved when humans store carbohydrates for later use. 2
- (iii) Use data in the table to assess the hypothesis made by the coach. 5

End of Question 30

Question 31 — The Chemistry of Art (25 marks)

(a) A solution of iron(II) sulfate is mixed with a solution of potassium permanganate under acidic conditions, resulting in a colour change.

(i) What is the observed colour change? **1**

(ii) Account for the colour change in terms of oxidation states. Include relevant half-equations in your answer. **3**

(b) The table provides examples of the colours shown by transition metal complexes.

<i>Metal ion</i>	<i>Ligand</i>	<i>Colour of complex</i>
Cu^{2+}	H_2O	light blue
Cu^{2+}	NH_3	dark blue
Cr^{3+}	H_2O	green
Cr^{2+}	H_2O	violet

(i) Account for the differences in colours shown by these transition metal complexes. **2**

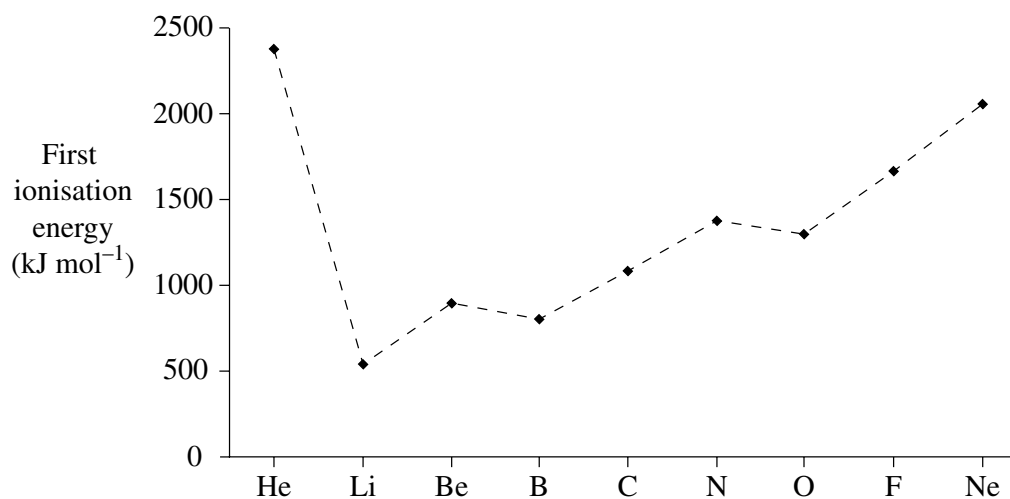
(ii) Assess the importance of minerals as pigments to TWO ancient cultures. **4**

(c) Describe the methodology and use of laser microspectral analysis, and assess the impact of advances in the understanding of emission spectra on the development of technologies. **7**

Question 31 continues on page 34

Question 31 (continued)

- (d) (i) Define the Pauli exclusion principle. **1**
- (ii) Distinguish between first ionisation energy and electronegativity. **2**
- (iii) The graph shows the first ionisation energies for the elements helium to neon. **5**



Analyse the relationship between first ionisation energy and electron configuration, using data from the graph.

End of Question 31

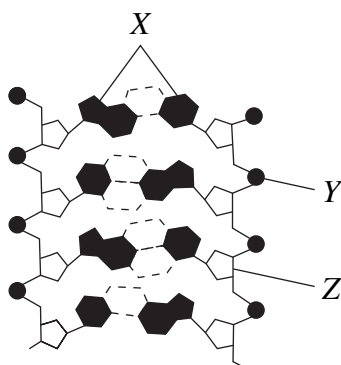
Question 32 — Forensic Chemistry (25 marks)

- (a) The diagram shows the emission spectra of three elements and an unknown mixture.



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- (i) Name TWO elements present in the unknown mixture. 1
- (ii) Describe how emission spectra assist in the identification of elements in a mixture. 3
- (b) The diagram illustrates the structure of DNA.



- (i) Identify the components labelled X, Y and Z. 1
- (ii) Explain how the analysis of DNA allows for identification of relationships between people. 2
- (iii) Discuss the ethics of maintaining data banks of DNA. 3

Question 32 continues on page 36

Question 32 (continued)

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|-----|--|----------|
| (c) | Describe the structure and composition of proteins, and assess the potential of electrophoresis as a method for their analysis. | 7 |
| (d) | (i) What type of carbohydrate is sucrose? | 1 |
| | (ii) Describe a chemical test that can be used to distinguish glucose from starch. | 2 |
| | (iii) A scientist wanted to develop a method of separation for a mixture of six carbohydrates using high performance liquid chromatography. The diagram shows the chromatograms obtained for the mixture using two different solvents as the mobile phase. | 5 |

Awaiting Copyright Clearance

Describe high performance liquid chromatography, and assess the effectiveness of the different solvents used by the scientist for separating the mixture of carbohydrates.

End of paper

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

[illegible]

Lanthanides

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm [144.9]	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89	Ac	Actinium	[227.0]
90	Th	Thorium	232.0
91	Pa	Protactinium	231.0
92	U	Uranium	238.0
93	Np	Neptunium	[237.0]
94	Pu	Plutonium	[244.1]
95	Am	Americium	[243.1]
96	Cm	Curium	[247.1]
97	Bk	Berkelium	[247.1]
98	Cf	Californium	[251.1]
99	Es	Einsteinium	[252.1]
100	Fm	Fermium	[257.1]
101	Md	Mendelevium	[258.1]
102	No	Nobelium	[259.1]
103	Lr	Lawrencium	[262.1]

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.

The atomic weights of Np and Tc are given for the isotopes ^{237}Np and ^{99}Tc .

**2005 HSC Notes from
the Marking Centre
Chemistry**

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2005 HSC NOTES FROM THE MARKING CENTRE

CHEMISTRY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It provides comments with regard to responses to the 2005 Higher School Certificate examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

This document should be read along with the relevant syllabus, the 2005 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

General Comments

In 2005, 10 083 candidates attempted the Chemistry examination. The most popular electives were Shipwrecks, Corrosion and Conservation (47%) and Industrial Chemistry (36%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

In 2005, at least one question in Section 1 Part B focused on the skills content in Module 9.1. Candidates who had actively planned and performed practical experiences clearly demonstrated a deeper knowledge and understanding of the content described in this module. In addition, there was clear evidence that many candidates had not performed the first-hand investigations. These are a crucial part of the syllabus and are examined. The knowledge gained from these experiences underpins the development of knowledge in Chemistry. In undertaking investigations involving secondary sources, candidates should check a variety of sources for reliability and accuracy of content, rather than rely heavily on one source.

Overall, the candidate's responses were appropriate and indicated a level of understanding of chemistry concepts appropriate for most HSC candidates. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with chemistry. This includes the writing of balanced equations, including states of matter, the mole concept and any calculations associated with it.

Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce clear and concise responses. This is particularly so in holistic questions which need to be logical and well structured. The use of dot points, diagrams and/or tables may be used to demonstrate their deeper level of understanding in Chemistry and avoid internal contradictions.

Better responses indicated that candidates are following the instructions provided on the examination paper. In these responses, candidates:

- show all working for numerical questions and express their answers with the correct number of significant figures
- think carefully about the units to be used and the quantities to be substituted into formulae.
- do not repeat the question as part of the response
- look at the structure of the whole question and note that in some questions the parts follow from each other ie responses in part (a) lead to the required response in part (b) etc
- use appropriate equipment, for example, pencils and a ruler to draw diagrams and graphs. (A clear plastic ruler would aid candidates to plot points that are further from the axes and rule straight lines of best fit.)

The option question is divided into a number of parts: candidates should clearly label each part of the question when writing in their answer booklets. In part ©, many candidates wrote a lot of information that was not relevant to the question. Some responses show evidence of rote learning an anticipated answer based on a single source. These did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Some candidates are responding to more than one option question: candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	B
2	C
3	B
4	D
5	B
6	A
7	C
8	A

Question	Correct Response
9	D
10	C
11	A
12	B
13	C
14	D
15	A

Part B

Specific Comments

Question 16

- (a) The majority of responses correctly named an appropriate alkene. Weaker responses named a gaseous alkene.
- (b) Responses that successfully linked the identified hazard and how it was addressed scored well. Responses that scored poorly did not identify a specific hazard for the first hand investigation and/or used vague or generic terms to outline how to address the hazard.

- (c) ©Better responses indicated the key elements of a safe, experimental procedure and identified appropriate reactants for this investigation. Weaker responses incorrectly included results and presented contradictory data.

Question 17

- (d) The majority of responses stated an appropriate reason for this difference.
- (e) Better responses clearly set out all relevant working and left the rounding off until the last step.

Question 18

Most responses were able to identify impacts of a biopolymer. Better responses indicated that candidates had gathered information from secondary sources about recent developments of a named biopolymer and included a judgement about its impacts. There was evidence that these responses were structured to include only the relevant information required by the question. Weaker responses could not distinguish between a polymer and a biopolymer.

Question 19

- (f) Candidates are reminded that their answer should identify a specific impact rather than offer a general statement, such as ‘the chemicals harm the environment’.
- (b) Better responses included balanced half equations or an overall equation and included identification of the anode, cathode and electrolyte.

Question 20

Better responses included balanced chemical equations and gave details about the reactants, products and conditions. They included labelled diagrams of appropriate laboratory equipment.

Question 21

Better responses showed a clear understanding of the chronological development of knowledge of acid-base theory and drew out and related the implications of each development. They used precise terms with respect to ions, atoms and molecules.

Question 22

- (a) Better responses identified the independent and dependent variables and labelled the axes correctly. They ensured that axes had linear scales that used the extent of the grid provided. Most candidates plotted the points correctly by marking the point with a cross or a circle. The better candidates used a pencil and ruler to draw two lines of best fit that intersected at a point and left out the outlier point from the line of best fit.
- (b) Most candidates identified the correct volume; however, many did not use the correct unit for volume, milliliter (mL), although it was indicated in the table. The better responses identified that hydrochloric acid (HCl) was the limiting reagent.

- (c) The better responses included the correct balanced equation, all the steps in their working, correct information from the data sheet and correct volume with correct units. They calculated the number of moles of zinc and used the correct number of significant figures.

Question 23

- (g) This part of the question was well answered. Most candidates were able to state that incomplete combustion results when there is a lack of oxygen.
- (h) This part of the question was poorly answered. Better responses included a balanced equation for the incomplete combustion of methane with correct states for each reactant and product. Weaker responses did not identify the products of incomplete combustion or gave an equation for the complete combustion of methane.

Question 24

- (a) This part of the question was extremely poorly answered by the majority of candidates, with the most common errors being the failure to include states, and recognition that carbon dioxide, water and an aqueous salt were formed.
- (b) Better responses accurately calculated the number of moles of acid.
- (c) Better responses calculated the moles of excess hydrochloric acid neutralised by the sodium hydroxide, then went on to calculate the mass of calcium carbonate.

Question 25

- (a) The better responses were able to clearly use the data to calculate the percentage of total dissolved solids. A significant number of candidates were unable to distinguish between the insoluble solids and the TDS.
- (b) The better responses described a distinguishing chemical test to identify lead (eg precipitation) and supported it with a correct chemical equation.
- (c) Generally well answered.

Question 26

Most responses described the processes involved in purifying the contaminated water. Better responses generally identified, described and made a judgement about each step of the process often using five to seven steps.

Question 27

- (a) Generally well answered.
- (b) (i) The better responses were able to show the initial spikes of increased concentration, the correct changes to each species concentration and the re-establishment of equilibrium. The majority of candidates did not include the initial spikes.

- (ii) The better responses were able to link the reduction in volume to an increase in pressure and use Le Chatelier's Principle to explain the shift to the side with least moles of gas.

Section II – Options

Question 28 – Industrial Chemistry

- (a)
 - (i) Most candidates correctly identified oleum.
 - (ii) Better candidates clearly linked the chemical properties of both dilute and concentrated sulfuric acid to transport safety requirements. Weaker responses included very general safety requirements such as the use of gloves, goggles etc which did not relate to the transport issues.
- (b)
 - (i) Better responses gave reasons in terms of the molar ratio of gas volume for the effects of the increase in pressure on the yield of dinitrogen tetroxide, and indicated that the addition of a catalyst would not affect the yield.
 - (ii) Better responses correctly determined the equilibrium concentrations of each species, and then substituted correctly into the equilibrium expression. They also accounted for the change in the equilibrium constant in terms of the exothermic nature of the forward reaction. Weaker responses did not include a correct equilibrium expression as part of their working.
- (c) Better responses included the advantages and disadvantages of the three methods of sodium hydroxide (NaOH) production, as well as descriptions of the changes in chemical processes that have occurred. These responses also included relevant chemical equations. Weaker responses were able to provide good descriptions of the advantages and disadvantages, but were limited in their understanding of the chemistry involved in the processes.
- (d)
 - (i) Most candidates were able to correctly identified the compound as sodium carbonate.
 - (ii) Better candidates went beyond simply processing information from the diagram and gave a clear overview, with relevant equations, of the chemistry of the ammonia recovery. Many candidates with poorer understanding of the chemistry confused reactions and gave incorrect reactants.
 - (iii) Better responses identified at least three chemical issues from the process, namely excess calcium chloride, thermal pollution and ammonia leakage and made judgements about the effectiveness of the treatment methods identified. Weaker responses outlined problems without linking to the Solvay process specifically, for example noise, dust, and quarrying.

Question 29 – Shipwrecks, Corrosion and Conservation

- (a) (i) This question was generally well answered.
- (ii) The better responses compared the same property for both named steels. Properties and uses had to be consistent with the type of steel. The more concise answers were presented in a table.
- (b) (i) The majority of candidates sketched in general terms the effect of one factor such as lower oxygen concentrations, lower temperature and the presence of anaerobic bacteria on the rate of corrosion.
- (ii) Very few candidates provided characteristics and features of the different chemical processes occurring in the beaker. The better responses gave detailed observations of the displacement reaction occurring on the magnesium strip and the corrosion processes occurring on the iron nail and included chemical equations when describing an experiment.
- (c) Better responses described at least three corrosion protection technologies including both cathodic protection and physical barriers, relating the technology to advances in chemistry, outlined advantages and disadvantages of each technology or advance, and supported their answer with chemical equations.
- (d) (i) This question was well answered.
- (ii) Most candidates addressed the purpose of placing the cannon in dilute sodium hydroxide.
- (iii) Better responses were carefully constructed and provided features and characteristics of the procedure, detailing the processes occurring at the anode and cathode. A labelled diagram and correct equations made their answers succinct. A significant number were able to justify either why electrolysis was used or particular processes occurring at the anode and cathode.

Question 30 – Biochemistry of Movement

- (a) (i) Better responses indicated an understanding of chemical structures, for example correctly drawing the covalent bonds of the oxygen of the hydroxyl group.
- (ii) Most candidates had a basic understanding of bonding within an alkanol with most recognising the importance of the role of hydroxyl groups in polarity and thus solubility in water or their importance in hydrogen bonding and thus viscosity. Few candidates were able to link the hydroxyl groups to hydrogen bonding and therefore to both viscosity and solubility.
- (b) (i) The majority of candidates were able to quantify the amount of adenosine triphosphate (ATP) produced by both processes.
- (ii) Most candidates showed knowledge of oxidation and reduction in terms of the loss and gain of electrons and that NADH and FADH₂ played a role in respiration. Better responses demonstrated an in-depth knowledge of glycolysis, the tricarboxylic acid cycle and oxidative phosphorylation and outlined precisely the role of NADH and

FADH₂. These responses linked the role of NAD⁺/NADH and FAD/FADH₂ to the production of ATP.

- (c) Better responses demonstrated knowledge of both protein structure and shape (primary, secondary, tertiary and quaternary) and the current model of enzyme function, ie the lock and key model. They provided evidence of a relationship between chemical features and the structure and shape and included the judgement necessary in the answer.

Most candidates showed knowledge of the lock and key model of enzymic function but were unable to show how this model was reliant on the specificity of shape of the active site on the enzyme. The better candidates were able to show how the shape could be changed through denaturation by pH change or temperature change. Very few candidates included bonding formed at the active site.

- (d) (i) The majority of candidates showed an understanding of the components of carbohydrates being carbon, oxygen and hydrogen. Better responses provided the general formula for carbohydrates as C_x(H₂O)_y.
- (ii) The majority of candidates were aware that carbohydrates were stored as glycogen. Better responses demonstrated knowledge of the process by which carbohydrates are stored, in particular the condensation reaction that needs to take place between the glucose monomers in order to make glycogen.
- (iii) The majority of candidates were able to draw information from the table to say that both the sports drinks contained carbohydrates and were needed for energy and make a judgement about the hypothesis. Better responses correctly discussed the metabolic processes occurring with the sprinter. They understood the immediate need to replace the used up ATP by the breakdown of creatine phosphate and not carbohydrates. Since neither of the sports drinks contain creatine phosphate they will not provide energy to sprinting athletes.

The majority of candidates were able to articulate the metabolic processes occurring with the marathon runner. The major error in this section was that most candidates had no idea about the time frame within which the metabolic process takes place and assumed that glucose would immediately produce energy once it had been drunk, in the form of the sports drinks.

Question 31 – Chemistry of Art

- (a) (i) This question was poorly done. There is evidence that candidates have not had first-hand experience of the use of potassium permanganate as a strong oxidising agent.
- (ii) Better responses demonstrated appropriate use of the Data Sheet. They stated reasons for oxidation state changes for both metals and linked the reactions.
- (b) (i) Better responses indicated precise relationships between different metals, different oxidation states, different ligands and different colours.
- (ii) Candidates generally had a good knowledge of the use and importance of pigments in different ancient cultures.

- (i) Many candidates could outline how laser microspectral analysis works; better responses gave a good explanation with clear reference to vaporisation and excitement of the vapour. They included the main features of emission spectrum formation and provided a judgement on the types of technologies used.
- (d)
 - (i) Most candidates have good knowledge of the Pauli Exclusion Principle.
 - (ii) Most candidates identified the difference between first ionisation energy and electronegativity.
 - (iii) Better responses referred to effective nuclear charge, and could explain the anomalies for boron and oxygen with reference to the sub-shells.

Question 32 – Forensic Chemistry

- (a)
 - (i) The majority of candidates correctly identified the elements.
 - (ii) Most candidates demonstrated some understanding of emission spectra. Better responses included a labelled diagram to describe the emission.
- (b)
 - (i) Most candidates could identify only 1 or 2 of the three labelled components. A significant number of candidates confused deoxyribose sugar and the phosphate groups.
 - (ii) Better responses clearly identified the types of DNA segments that are used for comparison between individuals and had a good understanding of introns and related these to similarities among individuals.
 - (iii) Better responses clearly identified issues relating to DNA data banks and provided specific detail of practices that were positive and negative for maintaining DNA databanks.
- (c) The better responses provided the features and characteristics of both proteins and electrophoresis. Their answers were well structured with diagrams and headings to ensure that they addressed all parts of the question. Their judgement of the potential of electrophoresis as an analysis method was clearly defined.
- (d)
 - (i) Majority of candidates correctly identified the type of carbohydrate.
 - (ii) Most candidates could identify the distinguishing test. The better responses described the test and also provided the expected results including original colours as well as colours of the products ie the colour change.
 - (iii) Most candidates provided an answer in general terms rather than specifically to high performance liquid chromatography (HPLC). The better responses clearly outlined specific detail on HPLC including diagrams. These candidates provided a judgement on the effectiveness of the solvents used in separating the mixture of carbohydrates provided in the question, referring specifically to the information in the question.

Chemistry

2005 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.2.5 Column 2 dp 1	H6
2	1	9.2.2 Column 2 dp 4, 5	H9
3	1	9.2.3 Column 3 dp 6	H7, H9, H10, H12, H13
4	1	9.2.2 Column 2 dp 3	H9, H10, H14
5	1	9.2.4 Columns 3 dp 1 9.2.4 Column 2 dp 1, 2, 5	H7, H8, H14
6	1	9.3.5 Column 2 dp 7 9.3.5 Column 3 dp 2	H9
7	1	9.3.2 Column 2 dp 4, 5 9.3.2 Column 3 dp 1	H7, H8, H14
8	1	9.3.3 Column 2 dp 2, 5 9.3.3 Column 3 dp 7	H10, H12
9	1	9.3.4 Column 2 dp 9, 5 9.3.3 Column 2 dp 6, 7	H8
10	1	9.3.1 Column 2 dp 2 9.3.4 Column 2 dp 6, 7	H8, H10, H12
11	1	9.4.5 Column 2 dp 2	H6, H7, H12
12	1	9.1 H12.2 (b) H11.2 (c), (d) H12.4 (e), (f), 9.4.5.3.1	H11, H12
13	1	9.4.4 Column 2 dp 9	H9, H13
14	1	9.4.5 Column 2 dp 1, 2	H12, H14
15	1	9.4.5 Column 2 dp 1	H8, H10, H14
Section I Part B			
16 (a)	1	9.2.1 Column 3 dp 2	H9
16 (b)	2	9.2.1 Column 3 dp 2	H9, H12
16 (c)	2	9.2.1 Column 3 dp 2	H11
17 (a)	1	9.2.3 Column 3 dp 6 Column 2 dp 7	H7, H9
17 (b)	2	9.2.3 Column 3 dp 6 Column 2 dp 7	H7, H12, H14

Question	Marks	Content	Syllabus outcomes
18	4	9.2.2 Column 3 dp 1 Column 2 dp 1	H4, H9
19 (a)	1	9.2.4 Column 3 dp 3	H4, H7
19 (b)	3	9.2.4 Column 2 dp 4, 5, 6 Column 3 dp 3	H7, H8, H12, H13
20	7	9.2.3 Column 2 dp 5, 6 Column 3 dp 4, 5 9.3.5 Column 2 dp 2, 4, 5, 6 Column 3 dp 1	H8, H9, H11, H13, H14
21	5	9.3.4 Column 1, Column 2 dp 1, 2 Column 3 dp 1	H1, H6, H8
22 (a)	3	H12.3 (c)	H12, H13
22 (b)	2	9.3.2 Column 2 dp 9	H10, H12, H14
22 (c)	2	9.3.2 Column 2 dp 9	H10, H12, H14
23 (a)	1	9.4.1 Column 2 dp 3	H8
23 (b)	1	9.2.1 Column 3 dp 1 Column 2 dp 1	H8, H10, H13
24 (a)	1	9.2.1 Column 2 dp 1 9.2.1 Column 3 dp 1	H8, H10, H13
24 (b)	3	9.3.4 Column 3 dp 3	H10, H12, H13, H14
24 (c)	1	9.3.4 Column 3 dp 3	H10, H12, H13, H14
25 (a)	2	9.4.5 Column 3 dp 1 9.4.1	H2, H10, H14
25 (b)	2	9.4.3 Column 3 dp 1	H4, H8, H11
25 (c)	2	9.4.5 Column 3 dp 1 9.4.3 Column 3 dp 2	H3, H8, H11, H14
26	7	9.4.5 Column 3 dp 3 Column 2 dp 2, 3, 4	H3, H4, H8, H14
27 (a)	2	9.4.2 Column 2 dp 2, 3	H8, H12, H14
27 (b) (i)	2	9.4.2 Column 2 dp 3, 7, 9 9.3.2 Column 2 dp 4	H8, H10, H13
27 (b) (ii)	2	9.4.2 Column 2 dp 3, 7, 9 9.3.2 Column 2 dp 4	H8, H10, H12, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 28 — Industrial Chemistry			
28 (a) (i)	1	9.5.2.3.1, 9.5.3.2.3	H6, H10
28 (a) (ii)	3	9.5.3.2.7, 9.5.3.2.8, 9.5.3.3.3	H7, H8, H12, H14.1
28 (b) (i)	2	9.5.2.1, 9.5.2.2.1	H8, H10, H14
28 (b) (ii)	4	9.5.2.2.1, 9.5.2.2.2, 9.5.2.2.3, 9.5.2.3.3	H7, H8, H10, H12, H13, H14
28 (c)	7	9.5.4.2.2, 9.5.4.2.3, 9.5.4.3.2	H3, H7, H8, H10, H14
28 (d) (i)	1	9.5.6.2.1	H8, H10
28 (d) (ii)	2	9.5.6.2.3, 9.5.6.3.2	H8, H10, H13
28 (d) (iii)	5	9.5.6.2.1, 9.5.6.2.4, 9.5.6.3.3	H7, H8, H14
Section II			
Question 29 — Shipwrecks, Corrosion and Conservation			
29 (a) (i)	1	9.6.2.2.1	H6, H8
29 (a) (ii)	3	9.6.2.2.2, 9.6.2.2.3, 9.6.2.3.3	H6, H12, H13, H14
29 (b) (i)	2	9.6.5 (all)	H6, H8
29 (b) (ii)	4	9.6.2.2.4, 9.6.2.3.2, 9.6.4.2.2	H8, H10, H12, H13, H14
29 (c)	7	9.6.4.2.1, 9.6.4.2.3, 9.6.4.2.4, 9.6.4.3.3, 9.6.4.3.4	H3, H7, H8, H10, H14
29 (d) (i)	1	9.6.7.2.5, 9.6.7.3.1	H8, H14
29 (d) (ii)	2	9.6.7.2.1, 9.6.7.2.5, 9.6.7.3.1	H8, H14
29 (d) (iii)	5	9.6.7.2.1, 9.6.7.2.3, 9.6.7.2.4, 9.6.7.2.5, 9.6.7.3.1	H8, H10, H13, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 30 — The Biochemistry of movement			
30 (a) (i)	1	9.7.3 Column 3 dp 1, 2	H9, H10, H13
30 (a) (ii)	3	9.7.3 Column 3 dp 2	H9, H14.1d
30 (b) (i)	2	9.7.7 Column 3 dp 1 9.7.1, 9.7.8 Column 2 dp 2	H7, H9, H12.4d, H14.1d
30 (b) (ii)	4	9.7.8 Column 3 dp 2 9.7.8 Column 2 dp 1, 2, 3, 4, 5	H7, H9, H12.4d, H13.1a, H14.1d
30 (c)	7	9.7.4 Column 2 dp 1, 2, 3, 4, 5, 7, 8 9.7.4 Column 3 dp 1, 3	H2, H9, H14
30 (d) (i)	1	9.7.2 Column 2 dp 1	H9, H10
30 (d) (ii)	2	9.7.2 Column 2 dp 2, 3 9.7.2 Column 3 dp 1	H7, H8, H9, H10
30 (d) (iii)	5	9.7.7 Column 1, 9.7.8 Column 1 9.7.10 Column 1	H7, H8, H9, H14.1a, H14.3c
Section II			
Question 31 — The Chemistry of Art			
31 (a) (i)	1	9.8.4 Column 2 dp 4	H8, H14.1d
31 (a) (ii)	3	9.8.4 Column 2 dp 3 9.8.4 Column 3 dp 3	H8, H10, H13.1d
31 (b) (i)	2	9.8.4 Column 2 dp 4 9.8.5 Column 2 dp 3, Column 3 dp 2	H6, H12.3c, H14.1b, H14.1h, H14.3b
31 (b) (ii)	4	9.8.1 Column 3 dp 1, 2 9.8.1 Column 2 dp 1, 2, 3, 7, 8	H1, H4, H8, H12.4d, e, H14.1a, e
31 (c)	7	9.8.2 Column 3 dp 3 9.8.2 Column 1, Column 2 dp 2, 8, 9	H3, H6, H14.1c
31 (d) (i)	1	9.8.3 Column 2 dp 1	H6
31 (d) (ii)	2	9.8.3 Column 2 dp 5, 7 9.8.3 Column 3 dp 1	H6, H7
31 (d) (iii)	4	9.8.3 Column 3 dp 1, 2 9.8.3 Column 2 dp 1, 2, 4, 5, 6, 7, 8	H6, H14.2a, H14.3a, b, d

Question	Marks	Content	Syllabus outcomes
Section II			
Question 32 — Forensic Chemistry			
32 (a) (i)	1	9.9.6 Column 2 dp 4, 5 9.9.6 Column 3 dp 2	H6, H12.3c, H14.1a
32 (a) (ii)	3	9.9.6 Column 2 dp 4, 5 9.9.6 Column 3 dp 2	H6
32 (b) (i)	1	9.9.4 Column 2 dp 1	H9, H10, H12.3c
32 (b) (ii)	2	9.9.4 Column 2 dp 2, 3	H4, H14.1g
32 (b) (iii)	3	9.9.4 Column 3 dp 1	H4, H14.3d
32 (c)	7	9.9.3 Column 2 dp 1, 2, 3, 4, 6 9.9.3 Column 3 dp 1, 5, 9.9.3.1	H9, H5, H14
32 (d) (i)	1	9.9.2 Column 2 dp 2	H9
32 (d) (ii)	2	9.9.2 Column 3 dp 2 9.9.2 Column 2 dp 1, 2, 3, 4	H9, H10
32 (d) (iii)	5	9.9.5 Column 2 dp 2 9.9.3 Column 2 dp 5 9.9.3 Column 3 dp 4	H9, H12.3c, H14.1a



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Section I, Part B

Question 16 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correctly names an alkene that could realistically be used	1

Question 16 (b)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
- Names a potential hazard related to the investigation - Provides main features of how the hazard was addressed	2
- Names a potential hazard related to the investigation OR - Correctly addresses a specific hazard related to the investigation	1

Question 16 (c)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
- Correctly outlines the procedure - Identifies an appropriate reagent - Identifies both appropriate organic substances - States use of small amounts	2
- Identifies an appropriate reagent OR - Correctly outlines procedure	1

Question 17 (a)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
States a correct reason	1

Question 17 (b)

Outcomes assessed: H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Calculates the correct mass of ethanol - Shows working	2
- Calculates the change in temperature correctly OR - Calculates the correct molar mass for ethanol OR - Correct substitution into formula	1

Question 18

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
- Names a recent development of a named biopolymer - States desirable properties of biopolymers - Identifies properties of biopolymers that allow an impact on society and environment - Makes a judgement about the impact of the biopolymer on both society and the environment	3–4
- Names a recent biopolymer - Identifies a property of a biopolymer that differentiates it from a synthetic polymer - States an impact on society or the environment	2
- Names a recent biopolymer OR - States a desirable property of a biopolymer (eg biodegradable) OR - States an impact on either society or the environment	1

Question 19 (a)

Outcomes assessed: H4, H7

MARKING GUIDELINES

Criteria	Marks
Identifies one environmental impact	1

Question 19 (b)

Outcomes assessed: H7, H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Identifies the composition of the anode, cathode and electrolyte - Provides two correct half-equations or one overall balanced equation	3
- Identifies the composition of the anode, cathode and electrolyte OR - Provides two correct half-equations or one overall balanced equation	2
- Identifies the composition of either the anode or cathode OR - Identifies the electrolyte	1

Question 20

Outcomes assessed: H8, H9, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of the chemistry of fermentation and esterification - Includes two correct balanced chemical equations - Describes procedures in each of three steps including at least one diagram	6–7
- Provides characteristics and features of the chemistry and procedure in fermentation and esterification - Includes one correct chemical equation - Describes procedures in two steps including at least one diagram	4–5
Provides characteristics and features of the chemistry and procedure in at least one step	2–3
Identifies one aspect of the chemistry or procedure in one step	1

Question 21

Outcomes assessed: H1, H6, H8

MARKING GUIDELINES

Criteria	Marks
- Describes the correct chronological development of theory of acids - Draws out the implications of each development - Provides at least three definitions of acids - Indicates limitations of earlier definitions	4–5
- Outlines correct chronological development of theory of acids - Provides two definitions of acids and indicates a limitation of one definition	3
Provides one definition of acids and indicates a limitation of that definition or implication of its development OR	2
States at least two definitions of an acid	
States one definition of an acid	1

Question 22 (a)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Outlier plotted but not included in graph (line of best fit) - Intersection point indicated - Lines connecting data points are straight - Points plotted correctly - Axes labelled - Linear scales used on axes	3
- Points plotted correctly - Axes labelled with units - Linear scale used on axes	2
- Axes labelled - Linear scale used on axes OR	1
Points plotted correctly	

Question 22 (b)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
States correct value and provides appropriate supporting evidence including reference to HCl	2
380 mL, without explanation	1

Question 22 (c)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Calculates volume of gas - Shows all relevant working	2
- Calculates moles of zinc OR - Writes correct balanced chemical equation OR - Provides an estimate of the volume of gas using the graph	1

Question 23 (a)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Identifies an appropriate circumstance	1

Question 23 (b)

Outcomes assessed: H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
Writes a correctly balanced equation	1

Question 24 (a)

Outcomes assessed: H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
- Writes a correctly balanced equation - Includes states of matter	1

Question 24 (b)

Outcomes assessed: H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the moles of hydrochloric acid	1

Question 24 (c)

Outcomes assessed: H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Calculates the mass of calcium carbonate and shows working and significant figures and units	3
Provides some correct steps in the calculation	2
- Calculates moles of NaOH OR - Writes balanced equation for the reaction of NaOH with HCl or correct mole ratio OR - Calculates molar mass of calcium carbonate	1

Question 25 (a)

Outcomes assessed: H2, H10, H14

MARKING GUIDELINES

Criteria	Marks
Calculates the percentage of total dissolved solids in the sample	2
Calculates the mass of total dissolved solids	1

Question 25 (b)

Outcomes assessed: H4, H8, H11

MARKING GUIDELINES

Criteria	Marks
Provides characteristics and features of a test that could be used to identify lead ions in sample	2
Names a correct reagent for identifying lead ions	1

Question 25 (c)

Outcomes assessed: H3, H8, H11, H14

MARKING GUIDELINES

Criteria	Marks
Provides a reason, with supporting argument, for monitoring a named ion	2
States a reason for monitoring a named ion	1

Question 26

Outcomes assessed: H3, H4, H8, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of two possible sources of contamination - Describes at least three methods that could be used to purify the water - Provides a judgement about the effectiveness of each of the methods used or a holistic judgement with supporting statement	6–7
- Provides characteristics and features of two possible sources of contamination - Describes two methods that could be used to purify the water - Provides a judgement in general terms about the effectiveness of at least one method used	4–5
- Outlines at least one possible source of contamination - Describes at least one method that could be used to purify the water	2–3
- Identifies one possible source of contamination OR - Identifies one method that could be used to purify the water	1

Question 27 (a)

Outcomes assessed: H8, H12, H14

MARKING GUIDELINES

Criteria	Marks
States that equilibrium has been reached	1

Question 27 (b) (i)

Outcomes assessed: H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
Correctly sketches all three trends	2
Identifies that equilibrium is re-established	
All three trends shown correctly	1

Question 27 (b) (ii)

Outcomes assessed: H8, H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Outlines how Le Chatelier's principle is applied to trends in the graph - States effect in terms of relative number of moles of gas on either side	2
- States Le Chatelier's principle OR - States that equilibrium is disturbed	1

Section II

Question 28 (a) (i)

Outcomes assessed: H6, H10

MARKING GUIDELINES

Criteria	Marks
Provides correct name	1

Question 28 (a) (ii)

Outcomes assessed: H7, H8, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of properties of both concentrated and dilute sulfuric acid - Relates properties of sulfuric acid to safety precautions necessary for its transport	3
Relates a property of concentrated or dilute sulfuric acid to a safety precaution necessary for its transport	2
- States a property of sulfuric acid OR - States a safety precaution necessary	1

Question 28 (b) (i)

Outcomes assessed: H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
- States the effect of the addition of a catalyst AND - States and gives reasons for the effect of an increase in pressure	2
- Correctly identifies the effect of the addition of a catalyst OR - Correctly identifies the effect of an increase in pressure	1

Question 28 (b) (ii)

Outcomes assessed: H7, H8, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Calculates the equilibrium constant for the reaction correctly OR - Shows working towards the K value - Provides a reason for the difference in K value related to temperature	4
- Calculates the equilibrium constant OR correctly substitutes into equation - Makes a statement in general terms about the difference in K value related to temperature	2–3
- Correct equilibrium expression OR - Makes a correct statement for the difference in K values	1

Question 28 (c)

Outcomes assessed: H3, H7, H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
- Provides a description of the three methods used for NaOH production identifying the improvements in technology that have taken place - Provides advantages and disadvantages of methods - Includes relevant chemical equations	6–7
- Provides a description of two methods for NaOH production using equations - Identifies some improvements in technology that have taken place OR - Discusses advantages and disadvantages of three methods with no chemistry	4–5
- Outlines two methods of NaOH production OR - Describes one method of NaOH production OR - Outlines advantage and disadvantage of two methods	2–3
- Names a process of NaOH production OR - Provides a correct statement about production of sodium hydroxide	1

Question 28 (d) (i)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
Correctly identifies X	1

Question 28 (d) (ii)

Outcomes assessed: H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
- Provides main features of chemistry of ammonia recovery - States a valid reason for recovery	2
- Provides main features of chemistry of ammonia recovery OR - States a valid reason for recovery	1

Question 28 (d) (iii)

Outcomes assessed: H7, H8, H14

MARKING GUIDELINES

Criteria	Marks
- Names at least three environmental issues arising from the process - Describes methods of dealing with these issues - Makes a judgement on their effectiveness	4–5
- Names two environmental issues arising from the process - Identifies methods of dealing with these issues	2–3
- Identifies one environmental issue arising from the process OR - Identifies one method of dealing with these issues	1

Question 29 (a) (i)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
Correct answer provided	1

Question 29 (a) (ii)

Outcomes assessed: H6, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Identifies similarities and/or differences in a specific property and states a use for each of two types of named steels	3
States a property (and a use) for each of TWO named steels	2
- States a property for each of two named steels OR - States a use for each of two named steels OR - States a property and a use of one named steel	1

Question 29 (b) (i)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
Outlines one factor that influences rate of corrosion of iron in deep water wrecks	2
Identifies one factor that influences rate of corrosion of iron in deep water wrecks	1

Question 29 (b) (ii)

Outcomes assessed: H8, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of two chemical processes including experimental observations OR - Outlines three chemical processes AND provides two full balanced redox equations or equivalent half equations	4
- Provides characteristics and features of one chemical process including experimental observations - Provides a balanced redox equation or equivalent half equations OR - Outlines two correct chemical processes - Provides a full balanced redox equation or equivalent half equations	3
- Outlines one chemical process OR - Provides two full balanced equations or equivalent half equations	2
- Identifies a chemical process OR - Describes an observation OR - Provides a correct half equation or balanced full redox equation	1

Question 29 (c)

Outcomes assessed: H3, H7, H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features three methods of corrosion protection - Relates advances in chemistry to the development of two of the three methods - Provides an advantage or disadvantage for all three methods - Includes relevant equations	6–7
- Provides characteristics and features two methods of corrosion protection (must include an equation) - Provides an advantage or disadvantage for each method	4–5
- Provides characteristics and features of one method of corrosion protection - Provides an advantage or disadvantage OR - Outlines two methods of corrosion protection - Provides one advantage or disadvantage	2–3
- Identifies a method of corrosion protection for ships OR - Makes a correct chemical statement about corrosion protection for ships	1

Question 29 (d) (i)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
Provides valid reason for the use of wax	1

Question 29 (d) (ii)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
- Provides reason for using NaOH solution - Provides reason for regular replacement of NaOH solution	2
- Identifies purpose of NaOH solution OR - Provides a reason for regular replacement of NaOH solution	1

Question 29 (d) (iii)

Outcomes assessed: H8, H10, H14, H13

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of the procedure - Provides characteristics and features of the processes occurring at the cathode and anode - Provides appropriate equations - Provides at least one supporting argument for using the procedure described	4–5
- Outlines the procedure - Outlines the processes occurring at cathode or anode	2–3
Provides one correct statement about procedure	1

Question 30 (a) (i)

Outcomes assessed: H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Provides the correct structure	1

Question 30 (a) (ii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Provides reason for the viscosity of glycerol - Provides reason for the solubility of glycerol in water	3
- Provides reason for the viscosity of glycerol OR - Provides reason for the solubility of glycerol in water	2
- Identifies that glycerol is a polar molecule OR - Identifies that glycerol can form hydrogen bonds with other molecules	1

Question 30 (b) (i)

Outcomes assessed: H7, H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
States the energy output of both glycolysis and respiration	2
- States the energy output of either glycolysis or respiration OR - States that energy is stored in the form of ATP	1

Question 30 (b) (ii)

Outcomes assessed: H7, H9, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of oxidation and reduction for respiration - Links the role of NAD^+/NADH and FADH_2/FAD to the production of ATP in respiration	3–4
- Outlines the role of oxidation and reduction in respiration (not all three parts) OR - Outlines the role of NAD^+/NADH and/or FAD/FADH_2 in respiration	2
Makes a correct statement about oxidation or reduction in respiration	1

Question 30 (c)

Outcomes assessed: H2, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Provides the relationship between the chemical features and the structure and shape of proteins - Describes the lock and key model of enzyme function - Provides a judgement about the importance of shape	6–7
- Provides the relationship between the chemical features and the structure and shape of proteins plus a judgement relating to shape - Describes the lock and key model of enzyme function and a judgement relating to shape	4–5
- Outlines how chemical features affect the structure and shape of proteins OR - Outlines the lock and key model of enzyme function	2–3
- Identifies one type of chemical feature that affects the structure or shape of proteins OR - Identifies that enzymes are catalysts OR - Identifies that proteins (or enzymes) are composed of amino acids	1

Question 30 (d) (i)

Outcomes assessed: H9, H10

MARKING GUIDELINES

Criteria	Marks
Provides a correct definition	1

Question 30 (d) (ii)

Outcomes assessed: H7, H8, H9, H10

MARKING GUIDELINES

Criteria	Marks
- Identifies that carbohydrates are stored as glycogen - Provides the features of the condensation reaction involved in glycogen formation, including that the starting material is glucose	2
- Identifies that carbohydrates are stored as glycogen OR - Identifies that glycogen is formed from glucose OR - Identifies that the chemical process is a condensation reaction	1

Question 30 (d) (iii)

Outcomes assessed: H7, H8, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies that the sports drinks contain carbohydrates that can provide energy - Compares metabolic pathways used by sprinters and/or marathon runners - Provides an assessment of the value of the two sports drinks - Identifies that the sodium in the sports drink will not provide energy	4–5
- Describes metabolic pathways used by sprinters or marathon runners OR - Identifies that both sports drinks contain carbohydrates which can provide energy - Provides an assessment of the value of at least one of the two sports drinks	2–3
- Identifies that the sports drinks contain carbohydrates that can provide energy OR - Identifies that sodium will not provide energy OR - Identifies sodium being involved as electrolytic replacement	1

Question 31 (a) (i)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
Identifies the observed colour change	1

Question 31 (a) (ii)

Outcomes assessed: H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
- States reasons for changes in colour for both manganese and iron - Gives both correct half-equations	3
- States change in oxidation state for both manganese and iron and gives one correct half-equation OR - Gives both correct half-equations	2
- States change in oxidation state for either manganese or iron OR - Gives one correct half-equation OR - States that either manganese is reduced or iron is oxidised	1

Question 31 (b) (i)

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
- States that colour of transition metal complexes depends on identity of metal ion and ligands - States that different oxidation states for the same metal exhibit different colours	2
- States that different oxidation states of transition metals exhibit different colours OR - States that colour depends on identity of metal ion and the surrounding ligands	1

Question 31 (b) (ii)

Outcomes assessed: H1, H4, H8, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of the use of minerals as pigments in two named ancient cultures - Provides a judgement of identified minerals/pigments in each of two named cultures	3–4
- Identifies two cultures and one mineral/pigment for each culture OR - Identifies one culture and at least two minerals/pigments	2
- Identifies two minerals/pigments used in ancient times OR - Identifies one mineral/pigment used by a culture	1

Question 31 (c)

Outcomes assessed: H3, H6, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of the methodology of laser microspectral analysis (LMA) - States one use of laser microspectral analysis (LMA) - Outlines an understanding of emission spectra formation - Provides a judgement on the types of technologies discussed	6–7
- Provides features and characteristics of the methodology of LMA - States one use of LMA - Provides a brief outline of emission spectra formation	4–5
- Outlines LMA - States one use of LMA - States one other technology or provides a brief outline of emission spectra formation	2–3
- Outlines emission spectra formation OR - Outlines one correct aspect of the methodology of LMA OR - States one use of LMA	1

Question 31 (d) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Gives the correct definition	1

Question 31 (d) (ii)

Outcomes assessed: H6, H7

MARKING GUIDELINES

Criteria	Marks
- States the definition of the first ionisation energy - States the definition of electronegativity	2
- States definition of the first ionisation energy OR - States definition of electronegativity	1

Question 31 (d) (iii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Explains trend across period two for first ionisation energy - Explains trend down noble gases for first ionisation energy - Explains trend anomalies for B and O	4–5
- Explains the trend across period two for first ionisation energy - Explains the difference in first ionisation energy between helium and lithium AND explains either anomaly at B or O OR - Explains both of the anomalies at B or O	2–3
States one trend related to the graph	1

Question 32 (a) (i)

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
Identifies the elements present in the sample	1

Question 32 (a) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of emission spectra and their use in identifying the elements in a mixture	3
Outlines the use of emission spectra in identifying the elements in a mixture	2
Provides one statement about atomic emission spectroscopy	1

Question 32 (b) (i)

Outcomes assessed: H9, H10, H12

MARKING GUIDELINES

Criteria	Marks
Identifies all the labelled components of DNA	1

Question 32 (b) (ii)

Outcomes assessed: H4, H14

MARKING GUIDELINES

Criteria	Marks
Provides reasons why DNA analysis is used to identify related people	2
Outlines DNA analysis	1

Question 32 (b) (iii)

Outcomes assessed: H4, H14

MARKING GUIDELINES

Criteria	Marks
Identifies two or more points relating to DNA databanks and provides points for and against their use	3
- Identifies an issue and provides points for and against the use of DNA databanks OR - Describes two issues relating to DNA databanks	2
Outlines an issue relating to DNA databanks	1

Question 32 (c)

Outcomes assessed: H5, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of the structure and composition of proteins - Describes the technique of electrophoresis using associated terminology - Provides a judgement about the effectiveness of electrophoresis	6–7
- Provides features and characteristics of the structure and composition of proteins - Outlines the technique of electrophoresis	4–5
- Outlines the technique of electrophoresis OR - Outlines the structure and composition of proteins	2–3
Provides one statement about proteins or electrophoresis	1

Question 32 (d) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correct type of carbohydrate identified	1

Question 32 (d) (ii)

Outcomes assessed: H9, H10

MARKING GUIDELINES

Criteria	Marks
- Identifies a correct chemical test and the compound it identifies - Describes the observations	2
Identifies a correct chemical test	1

Question 32 (d) (iii)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Provides the characteristics and features of high performance liquid chromatography (HPLC) - Makes a judgement about the effectiveness of the solvents	4–5
- Outlines HPLC - Makes a generalisation about the effectiveness of one solvent (A or B)	2–3
- Outlines a feature of HPLC OR - Makes a generalisation about the effectiveness of one solvent (A or B)	1